SESSION S01

Unmanned Aircraft: What They Are, What They Are For

Robotics, Drones & Autonomous Systems
KAUST Gifted Student Program · Summer Enrichment 2026

First powered flight on another planet 0:43

RQ-4 Global Hawk · U.S. Air Force, public domain

THE COURSE

Course structure

Where this session sits

Choose a class	Size a multirotor	Assemble	Commission	Simulate
S01 · Introduction	S03 · Multirotor theory	S05 · Structure	S07 · Finish the build	S09 · Mission Planner & SITL
S02 · Mission → the choice	S04 · Components & budget	S06 · Power & comms	S08 · Firmware & testing	S10 · Full mission & scripting

Three projects, not one. **In S01 and S02 you choose an aircraft class** for a problem with a real problem. **S03 and S04 are a separate project on multirotors** — how they are laid out, and what they are made of. **In S05–S10 you assemble a real one, commission it, and fly it in simulation.**

Session outline

- **Defining the unmanned aircraft**
  The terminology, and the four elements that make up a system.
- **How they are operated**
  Remote piloting, automation and autonomy — and the airspace rules.
- **Operational applications**
  Ten aircraft doing ten real jobs, defence and civil.
- **Mechanisms of sustained flight**
  The airfoil, the two governing facts, and the four configuration classes.
- **Exercise, then discussion**
  Four missions. You select a configuration for each.

PART ONE

Defining the unmanned aircraft

Four terms, and how they differ

Drone	EVERYDAY	Everyone knows what you mean, and it tells you nothing about what the thing is or does
UAV	UNMANNED AERIAL VEHICLE	**The aircraft, and only the aircraft.** The thing that leaves the ground
UAS	UNMANNED AIRCRAFT SYSTEM	The aircraft *plus* everything it needs to be useful. This is the word engineers should use
RPAS	REMOTELY PILOTED	A UAS with a qualified human pilot in the loop. The term regulators write rules about

> "How much does the drone weigh?" is an **ambiguous question** until you say whether you mean the vehicle or the system. You will be asked it in S04, and the answer is worth marks.

The four elements of an unmanned aircraft system

01

The air vehicle

Airframe, motors, battery, autopilot, sensors. The part everyone photographs.

02

The ground station

Where the mission is planned, watched and changed. A laptop, a tablet, or a shipping container.

03

The link

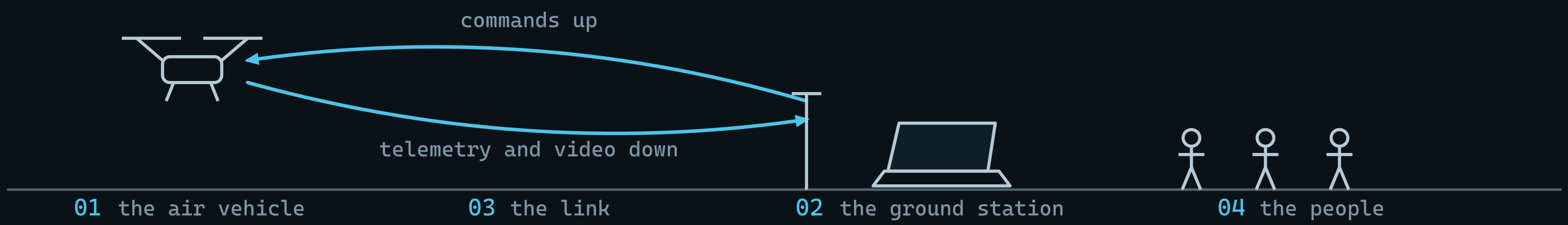
Radio between the two. Range, bandwidth and what happens the moment it drops.

04

The people

A pilot, usually an observer, and whoever owns the airspace you are in.

Three of those four are not the aircraft — and in the field, **that is where most of the trouble comes from.** A flat ground-station battery cancels a sortie exactly as completely as a broken propeller.



A QUESTION FOR THE ROOM

Which of those four fails most often in real operations?

Commit to one before any discussion. **An incorrect answer costs nothing here** — the same misjudgement costs a session at the bench in S08.

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60 SECONDS **Indicate your choice, then justify it in one sentence**

PART TWO

How they are operated

Remote piloting, automation and autonomy

01

Remote-piloted

A human moves the sticks. The autopilot holds attitude and altitude so the human does not have to. Almost all recreational flying is this.

02

Automatic

The aircraft flies a plan somebody wrote in advance — a list of waypoints, a survey grid, a return-to-launch. It executes. **It does not decide anything.**

03

Autonomous

The aircraft chooses *what to do next* from what it senses. Search until you find something, then change the plan because of what you found.

> Most products sold as "autonomous drones" are **automatic**. It is a marketing word far more often than an engineering one — and knowing the difference is most of what separates you from the brochure.

In S09 you will plan an automatic mission. In S10 you will command that same mission from a script. The distance between those two is the whole back half of this course.

Airspace and regulatory constraints

REGISTRATION	Above a mass threshold, the aircraft is registered — like a car
LICENCE	The pilot holds a certificate. It is a person who is qualified, not the aircraft
LINE OF SIGHT	You can usually only fly what you can see, unless you hold a specific approval
ALTITUDE	A ceiling, typically low, to keep you away from everything else that flies
PEOPLE	Not over crowds, and not over anyone who has not agreed to be there
PERMISSION	Somewhere, someone has to say yes. Near an airport, that is not a formality

Details differ by country. **Nothing in this course is flown outdoors** — every mission you plan is executed in simulation — but the shape of the rules is the same almost everywhere. Watch for what it costs in the next section: **one of the aircraft you are about to see flies from a single fixed depot**, and this is why.

PART THREE

Operational applications

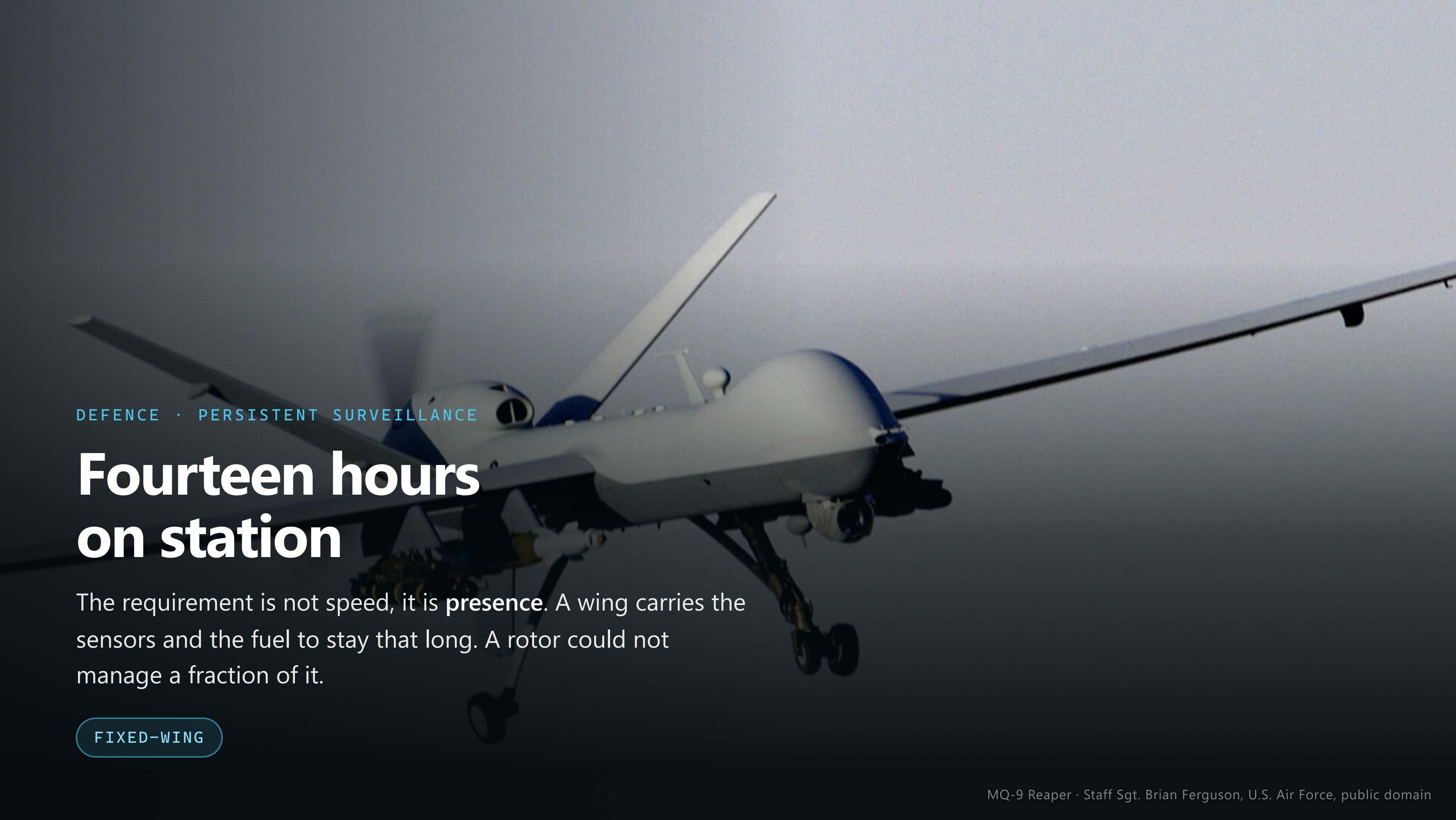
DEFENCE · STRATEGIC RECONNAISSANCE

Thirty hours of endurance on a single sortie

A forty-metre wing at eighteen kilometres, covering a country in a day. Endurance like that has only ever been a **wing** problem — and this one takes off and lands on a runway, like an airliner.

FIXED-WING

RQ-4 Global Hawk · U.S. Air Force, public domain



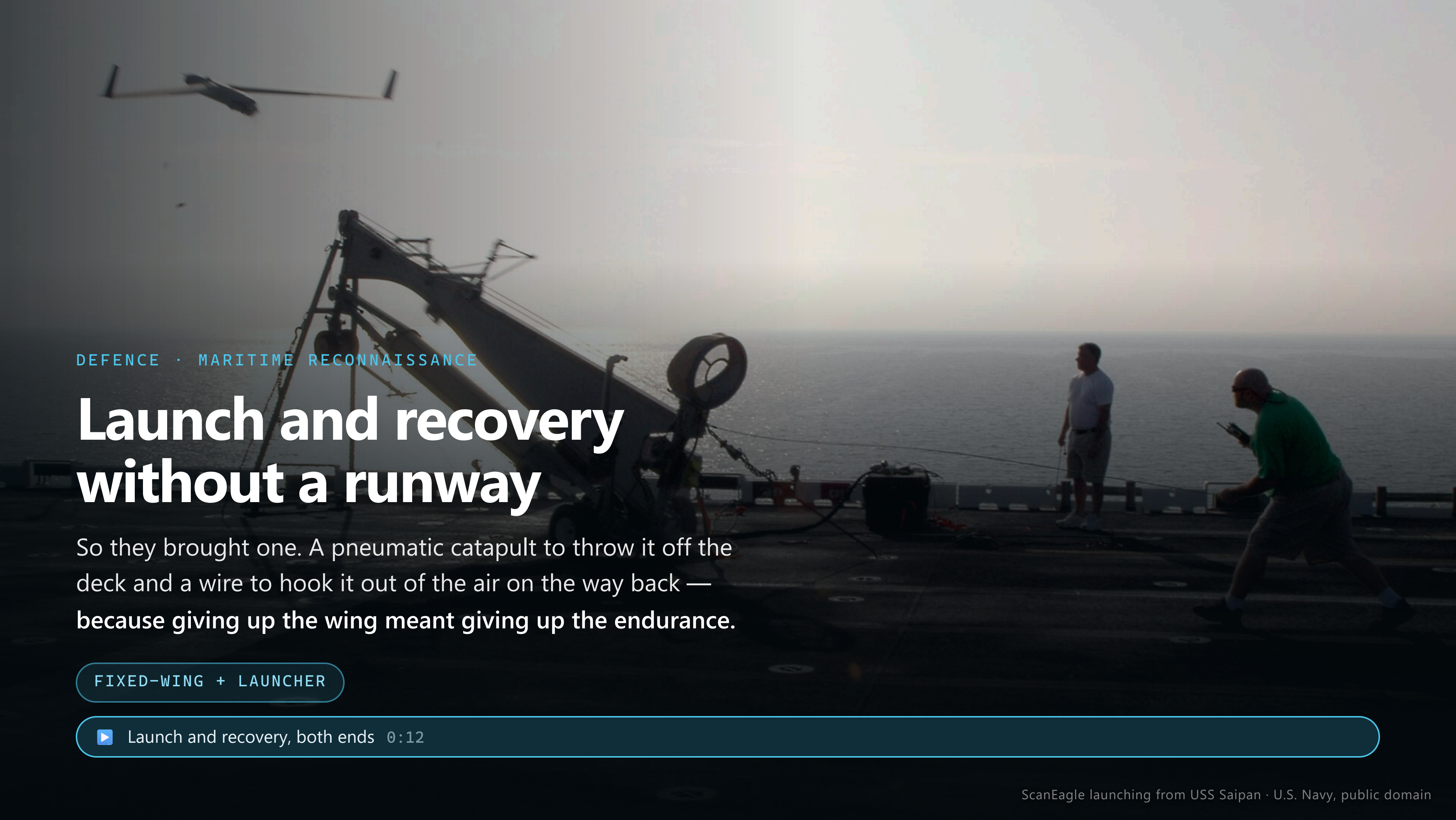
DEFENCE · PERSISTENT SURVEILLANCE

Fourteen hours on station

The requirement is not speed, it is **presence**. A wing carries the sensors and the fuel to stay that long. A rotor could not manage a fraction of it.

FIXED-WING

MQ-9 Reaper · Staff Sgt. Brian Ferguson, U.S. Air Force, public domain



DEFENCE · MARITIME RECONNAISSANCE

Launch and recovery without a runway

So they brought one. A pneumatic catapult to throw it off the deck and a wire to hook it out of the air on the way back — **because giving up the wing meant giving up the endurance.**

FIXED-WING + LAUNCHER

▶ Launch and recovery, both ends 0:12

ScanEagle launching from USS Saipan · U.S. Navy, public domain

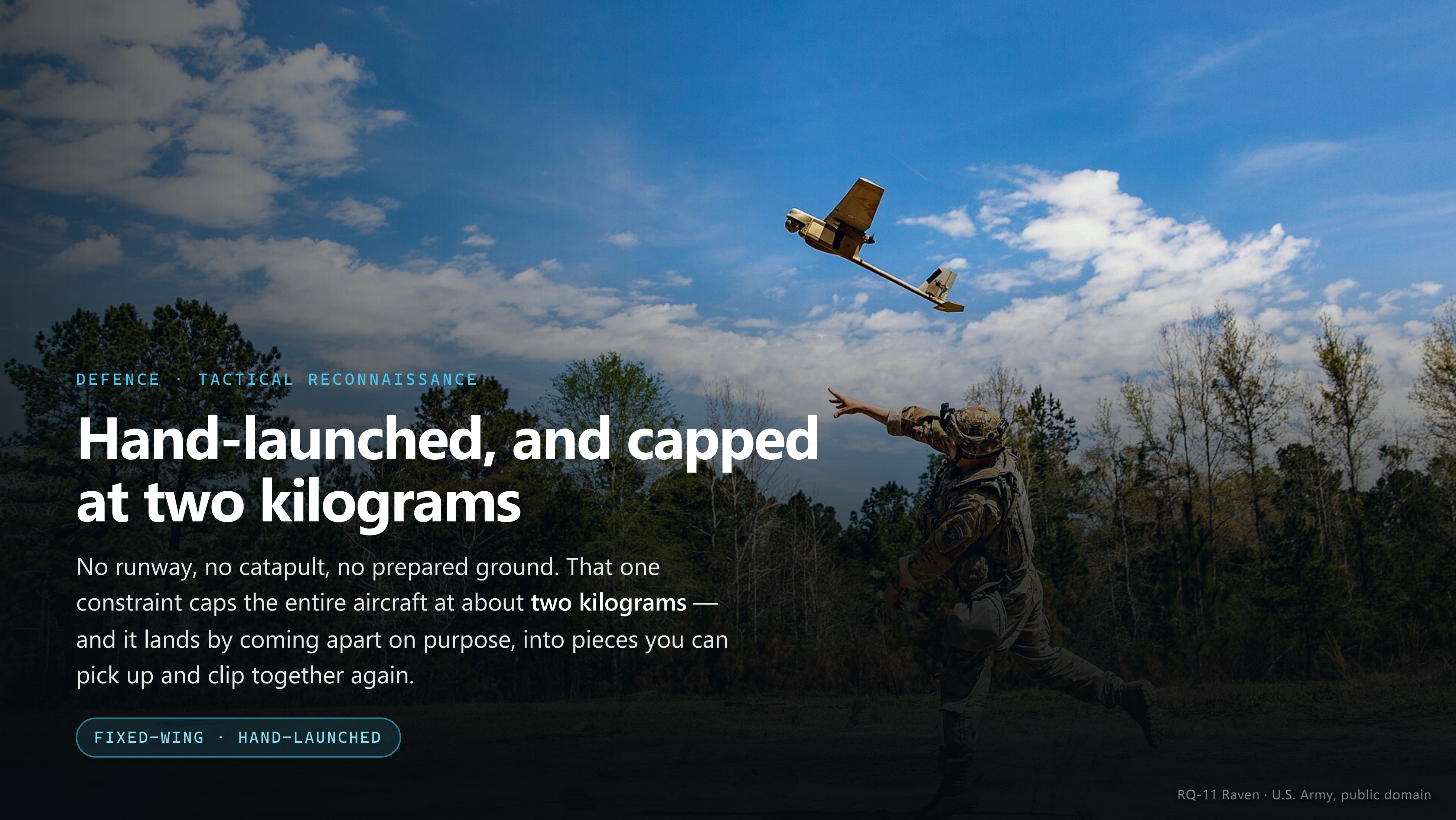
DEFENCE · MARITIME RECONNAISSANCE, ROTARY-WING

The same requirement, met with a rotorcraft

Same ship, same job, opposite answer. A rotorcraft needs no catapult and no net. It pays for that in endurance — and in a gearbox that has to be maintained at sea.

HELICOPTER

MQ-8B Fire Scout alongside USS McInerney · U.S. Navy, public domain



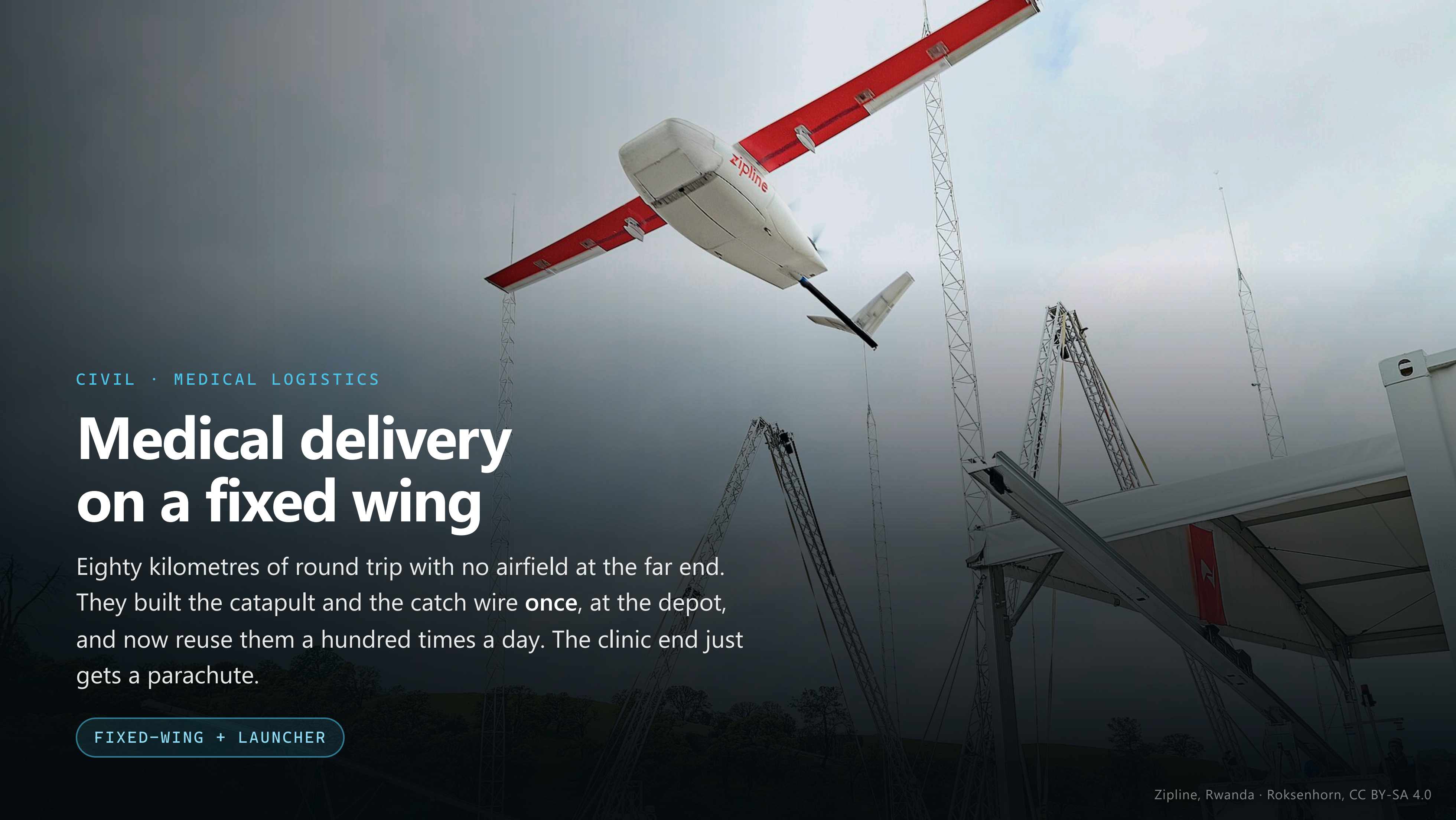
DEFENCE · TACTICAL RECONNAISSANCE

Hand-launched, and capped at two kilograms

No runway, no catapult, no prepared ground. That one constraint caps the entire aircraft at about **two kilograms** — and it lands by coming apart on purpose, into pieces you can pick up and clip together again.

FIXED-WING · HAND-LAUNCHED

RQ-11 Raven · U.S. Army, public domain



CIVIL · MEDICAL LOGISTICS

Medical delivery on a fixed wing

Eighty kilometres of round trip with no airfield at the far end. They built the catapult and the catch wire **once**, at the depot, and now reuse them a hundred times a day. The clinic end just gets a parachute.

FIXED-WING + LAUNCHER

Zipline, Rwanda · Roksenhorn, CC BY-SA 4.0

CIVIL · POWER LINE INSPECTION

Station-keeping beside a live conductor

The job is not covering ground. It is **holding a position** long enough for an inspector to decide what they are looking at, and then moving three metres and doing it again. Nothing with a wing can do that.

HELICOPTER

Camcopter S-100 · Schiebel Group, CC BY-SA 4.0

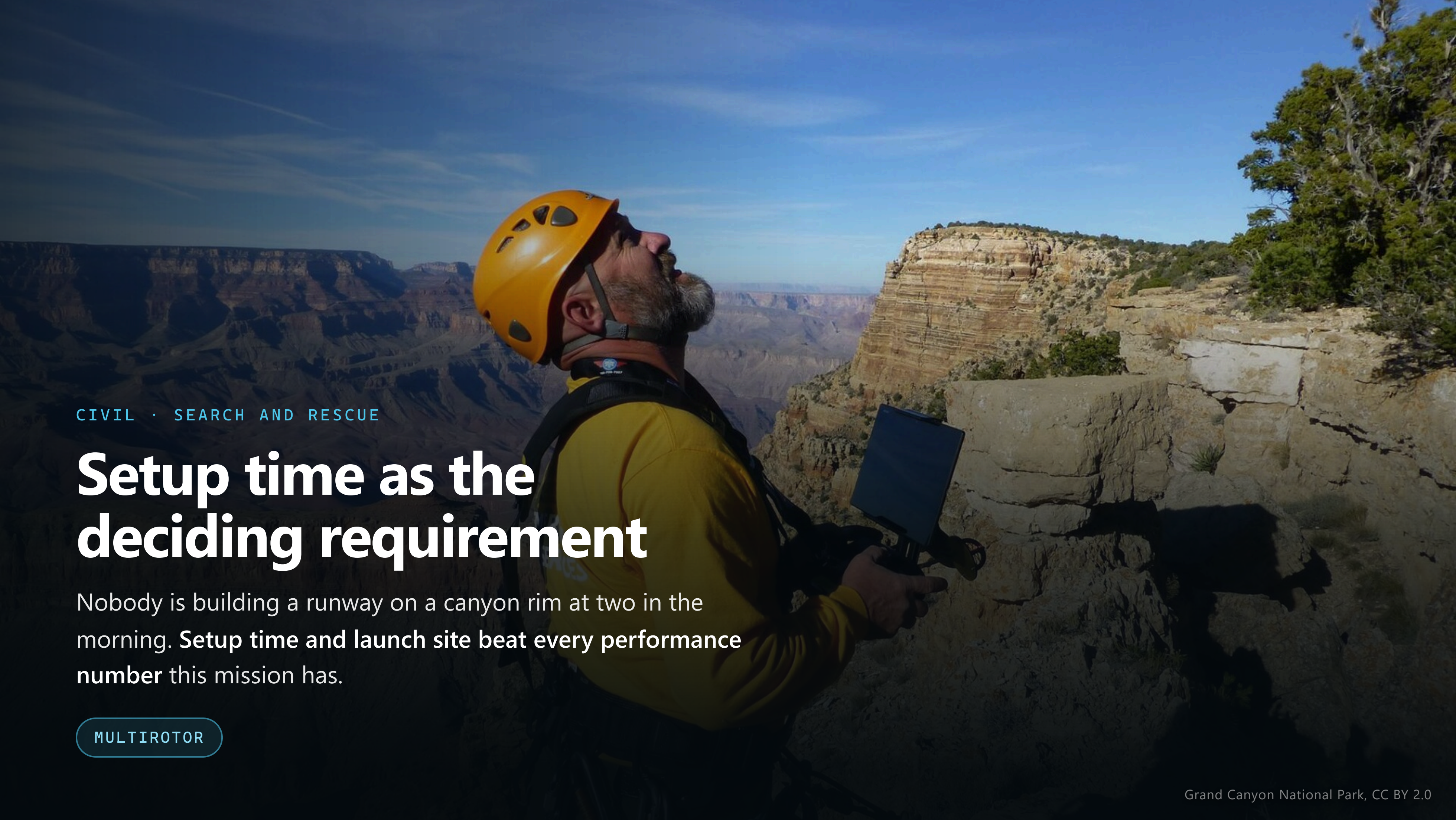
CIVIL · PRECISION AGRICULTURE

Spraying passes that require a full stop

A heavy tank over a small field, on a flight path that ends every pass with a full stop and a reversal. A wing would have to fly a circuit to come back. A rotor just turns around.

MULTIROTOR

Rice crop spraying · Christopher Hedreyd, PIA CALABARZON, public domain



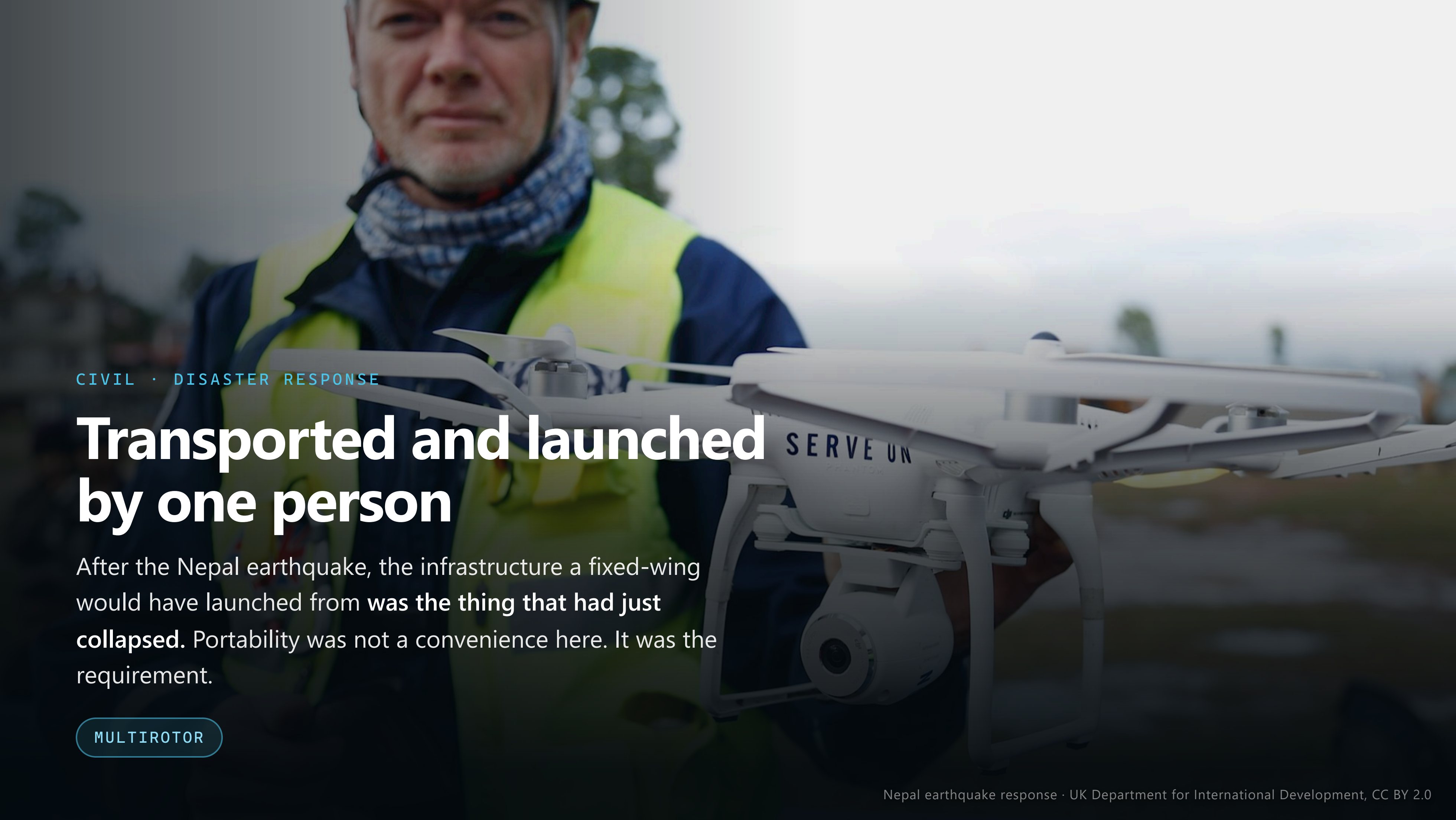
CIVIL · SEARCH AND RESCUE

Setup time as the deciding requirement

Nobody is building a runway on a canyon rim at two in the morning. **Setup time and launch site beat every performance number** this mission has.

MULTIROTOR

Grand Canyon National Park, CC BY 2.0



CIVIL · DISASTER RESPONSE

Transported and launched by one person

After the Nepal earthquake, the infrastructure a fixed-wing would have launched from **was the thing that had just collapsed.** Portability was not a convenience here. It was the requirement.

MULTIROTOR

Nepal earthquake response · UK Department for International Development, CC BY 2.0

Ten applications, three governing questions

01

Where does it launch from, and what catches it coming back?

The Global Hawk gets a runway. The ScanEagle brings a catapult. The Raven gets somebody's arm. Every one of those is a *site* constraint — and it decided the aircraft before any performance number was considered.

02

Does the job need the aircraft to stop?

Inspection and search do. Delivery and wide-area survey do not. That single question splits the configurations cleanly in half, and it is answered by the mission, never by the engineer.

03

What happens if you lose it?

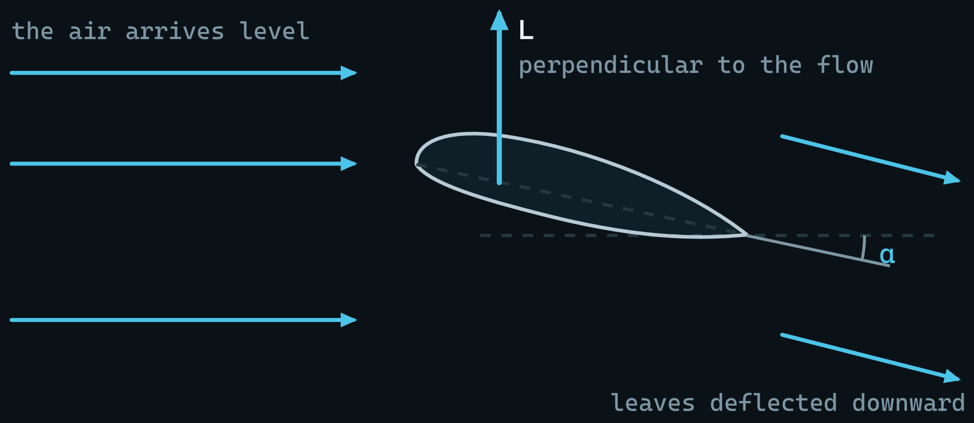
An airframe you accept losing stops needing a recovery system at all — and that changes every other line of the design. An airframe you cannot afford to lose buys redundancy with mass.

> Not one of those three is about top speed, endurance or payload. Those matter — but they come **later**, and you cannot size an aircraft until you know what kind of aircraft it is.

PART FOUR

Mechanisms of sustained flight

How an airfoil generates lift



The chord sits at **α** to the flow. The section **turns the air downward**, the air pushes back, and the part of that reaction across the flow is **lift**.

▶ Cambridge smoke tunnel — the flows do not meet 1:13

Nothing is sucked upward. **A wing throws air down**, and Newton's third law does the rest — the same trick a rotor uses.

$$C_L \approx 2\pi\,\alpha$$

C_L Lift coefficient — **not** a constant

α Angle of attack, in **radians**

The angle sets C_L — 5° gives ≈ 0.55 — and that is the C_L in **$L = ½\rho V^2 S \cdot C_L$**. Past about **15°** the flow separates and lift collapses: the **stall**, an *angle* before it is a speed.

The two halves of the flow never meet again. Equal transit time is **not true** — the upper flow arrives *first*, and a flat plate at an angle flies perfectly well.

Two governing facts

HOVERING IS EXPENSIVE

A rotor holds you up by throwing air downwards, continuously, for as long as you are up there.

$$P_{\text{hover}} \propto \frac{W^{3/2}}{\sqrt{A}}$$

P power to hover · **W** weight · **A** total disc area

Heavier costs **more than proportionally**. Bigger discs cost less. Double the disc area and you save about **30%** of the hover power — at the same weight.

A WING IS NOT

A wing holds you up by moving forward. You stop paying for weight and start paying only for **drag**.

$$L = \tfrac{1}{2}\,\rho V^2 S\, C_L \qquad T = \frac{W}{L/D}$$

L lift · **ρ** air density · **V** airspeed · **S** wing area · **C_L** lift coefficient · **T** thrust · **L/D** lift-to-drag ratio

A lift-to-drag ratio of 12 means **one twelfth of your weight in thrust** keeps you up. A rotor gets no such discount.

A wing covers far more ground per joule; a rotor can stop. **Every configuration argument is those two facts in tension.**

The four configuration classes

Multirotor

4+ FIXED-PITCH ROTORS
THRUST SET BY RPM

- \+ Hovers, holds a point, lands on anything flat
- – Everything it does, it does on hover power

Fixed-wing

A WING, IN FORWARD
FLIGHT ONLY

- \+ Range and endurance nothing else comes near
- – Cannot stop. Needs a way up and a way down

Helicopter

ONE LARGE ROTOR
VARIABLE PITCH

- \+ Best hover of any rotorcraft, and it cruises too
- – Mechanically complex, costly, maintained constantly

VTOL hybrid

A WING *PLUS* LIFT
ROTORS OR TILTS

- \+ Wing range with vertical launch and recovery
- – You carry both systems and tune two flight regimes

Four *classes*, not four products. Every aircraft in the last ten slides was one of these.

Watch a VTOL transition 2:12

PART FIVE

Exercise — 25 minutes

Four missions: select a configuration for each

MISSION 1 · FILM

A crew wants a **six-minute continuous shot** following a car along a coast road at 80 km/h, then climbing away into a wide static view of the bay.

MISSION 2 · PIPELINE

Every kilometre of a **400 km gas pipeline**, photographed twice a year at 5 cm/pixel. Access roads run alongside for most of its length.

MISSION 3 · PORT

The **underside of every crane boom** in a container port checked for cracks, close up, without stopping container movements.

MISSION 4 · RESEARCH

Three kilograms of samples moved 40 km from a remote valley station to a road head, twice a week, all year, in whatever weather turns up.

None of these is your mission brief. Those come in S02.

Task

CORE · EVERYONE FINISHES THIS

STRETCH · IF YOU FINISH EARLY

Twenty-five minutes. Nobody is marking this — it is the reasoning you will need in S02, on a mission that *is* marked.

Task

CORE · EVERYONE FINISHES THIS

- For each of the four, name **the configuration class** you would fly

STRETCH · IF YOU FINISH EARLY

Twenty-five minutes. Nobody is marking this — it is the reasoning you will need in S02, on a mission that *is* marked.

Task

CORE · EVERYONE FINISHES THIS

- For each of the four, name **the configuration class** you would fly
- For each one, name the **single constraint** that decided it — one sentence, no more

STRETCH · IF YOU FINISH EARLY

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Task

CORE · EVERYONE FINISHES THIS

- For each of the four, name **the configuration class** you would fly
- For each one, name the **single constraint** that decided it — one sentence, no more
- Where you rejected an obvious alternative, say what killed it

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STRETCH · IF YOU FINISH EARLY

- One of the four is genuinely arguable. Find it, and say what extra information would settle it

Twenty-five minutes. Nobody is marking this — it is the reasoning you will need in S02, on a mission that *is* marked.

Task

CORE · EVERYONE FINISHES THIS

- For each of the four, name **the configuration class** you would fly
- For each one, name the **single constraint** that decided it — one sentence, no more
- Where you rejected an obvious alternative, say what killed it

STRETCH · IF YOU FINISH EARLY

- One of the four is genuinely arguable. Find it, and say what extra information would settle it
- For that one, is "two aircraft" a better answer than any single aircraft?

Twenty-five minutes. Nobody is marking this — it is the reasoning you will need in S02, on a mission that *is* marked.

Discussion — two minutes per team

Take one mission per team and tell the room:

1. Which class you picked
2. The one constraint that decided it
3. What you rejected, and what killed it
4. What would change your mind

> Question 4 is the one that matters. An engineer who cannot say what would change their mind has not made a decision — **they have expressed a preference.**

NEXT SESSION

S02 — Mission Analysis

In S02 you stop looking at other people's problems and get one of your own. Three teams, three distinct problems, three very different environments.

> You now know what the shapes are and what they are for.
> **Next you find out what you have been asked to do.**

Nobody gets the same brief. That is deliberate.

IMAGE CREDITS

Photographs

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MQ-9 Reaper — Staff Sgt. Brian Ferguson, U.S. Air Force, public domain

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RQ-11 Raven — U.S. Army, public domain

MQ-8C Fire Scout — U.S. Navy, public domain

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